

MEMORANDUM
DEPARTMENT OF FISH AND GAME

STATE OF ALASKA
Division of Commercial Fisheries
Management and Development

TO: Yukon Staff
(See Distribution)

DATE: November 29, 1993

FILE NO: MEMOS\REGIONAL\
STF11-29.93

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SUBJECT: 1994 Fall Chum
Salmon Run Projection

This memorandum presents the 1994 fall chum salmon run projection which is based upon estimated spawner-return relationships and age composition data. The 1994 projected return is 605,151 fall chum salmon. This projection includes an estimated 112,000 age-5 shortfall from the 1989 brood year based upon performance of returning age-4 fish in 1993; a recruitment rate of only 0.8. Should the poor levels of production exhibited by the 1988 and 1989 brood years typify that of 1990, then a projection of 605,000 fall chums is highly optimistic.

HARVESTS

Annual harvest data presented in Table 1 for the years 1974-1977 were taken from respective Annual Management Reports (AMR's), while that for the years 1978-1991 are from Appendix A.20 of the 1991 AMR (unless otherwise indicated). The 1992 harvest data are from the 1992 AMR. All 1993 data are preliminary.

Adjustments to harvest data include the estimated unreported catch from Districts 5 and 6 in 1987 (approximately 115,800 fish) and are reflected under the subsistence catch for that year in Table 1. The District 6 department test wheel catches of fall chum salmon each year from 1988 through 1992 are included under the commercial catch figures for those years in Table 1. No test fish sales were made in 1993.

Although the reported Alaska subsistence harvest estimates for 1974-1978 may have been low due to timing of surveys, reported catches for those years are shown and used in the attached tables and analysis. While a limited U.S. subsistence fishery was prosecuted in 1993 prior to the complete closure, it is known that some level of illegal activity also occurred. I subjectively estimated the 1993 U.S. subsistence catch at 100,000 fall chum salmon, a level approaching the

1992 estimated harvest (111,109). I assumed, for the purpose of estimating total 1993 run size, that this level of harvest also accounted for illegal catches in the "protest fishery" as well.

TOTAL RUN SIZE AND 1993 ESCAPEMENT ESTIMATES

Annual Yukon River fall chum salmon run abundance is also shown in Table 1 and was estimated as twice the 4-area escapement index plus documented commercial and reported subsistence harvests for each year (Figure 1). Estimated annual returns during the 20-year period 1974-1993 were cyclic, with small returns realized in even-numbered years and larger returns in odd-numbered years (Figure 2). During that period, estimated total fall chum return in **odd-numbered** years ranged from approximately 353,000 (1993 - lowest return on record) to 1,625,000 (1975), averaging 950,000. In **even-numbered** years, total return estimates ranged from approximately 396,000 (1992) to 739,000 (1974), averaging 577,000. However, during the past 14-year period (1980-1993) estimated total run size has averaged only 814,000 in **odd** years and 566,000 in **even** years.

As with past projections, I once again evaluated total Yukon River fall chum escapements based upon observations to the Delta, Toklat, and Sheenjek Rivers in Alaska, and the Fishing Branch River in Canada (Table 2 and Figure 2). The 1993 minimum escapement goals to these four streams were 11,000, 33,000, 64,000, and 50,000, respectively. The combined 4-area index minimum escapement goal is 158,000 fall chum salmon. In addition to estimates of total escapement to the above four streams, annual estimates of border passage and subsequent spawning escapement also exist for Canadian fall chum stocks in the upper mainstem Yukon River (See Table 2). The minimum goal for those combined stocks is 80,000 fall chum spawners, i.e., border passage less harvest.¹

The preseason projected return of fall chum salmon to the Yukon River in 1993 was 734,000 fish. That projection included an anticipated drainage-wide shortfall of 80,000 age-5 fish from the 1988 brood year based upon performance of returning age-4 fish observed in 1992. Beyond the age-5 shortfall, it was anticipated that the age-4 component of Canadian stocks, particularly those of the mainstem, might also exhibit some weakness based upon the observed escapements in 1989. However, there was no reason to suspect a weakness in the return of age-4 fish to areas in the Alaskan portion of the drainage as escapement goals were achieved in the Sheenjek, Delta, and Toklat Rivers in 1989.

Near the estimated midpoint of the 1993 run, the department made a revised projection of total return of 350,000-500,000 fall chums based upon mean cumulative CPUE in lower river test

¹The U.S. and Canada has negotiated a twelve year rebuilding plan, beginning in 1990 and ending after the 2001 season, for Yukon River mainstem fall chum salmon. The objective of the plan is to rebuild the stock by achieving a spawning escapement of 80,000 or more fall chum salmon for all (four) brood years by the year 2001. The plan will endeavor to rebuild the stronger parent years in four years (one cycle) and the weaker parent years in twelve years (three cycles) in equal increments. The 1994 targeted level of escapement was > 66,000 fish.

nets and the passage estimate at Pilot Station. Additionally, test fish age composition data indicated that the proportion of age-4 fish was well below that projected preseason, suggesting poor recruitment from the 1989 parent year. Consequently, the fall season commercial fishery was not opened, and by mid-August, the sport and personal use fisheries were closed with restrictions imposed upon the Alaskan subsistence fishery.

By the end of August, it was apparent the Yukon River fall chum salmon run was extremely weak and another inseason projection of total return was made at 300,000-350,000 fish. This was based upon lower river test fish CPUE and age composition data, Pilot Station sonar passage estimate, and Sheenjek River spawning escapement estimate. As a result, subsistence salmon fishing within the Alaskan portion of the drainage was closed until further notice.

Evaluation of escapements in the Porcupine River drainage was measured by observations made in the Sheenjek and Fishing Branch Rivers. The sonar estimated escapement in the Sheenjek River in 1993 was approximately 43,000 fish for the period August 8 through September 28, 33% below the escapement goal of 64,000. The project operated for 52 days in 1993, whereas the next longest running operations were in 1991 (47 days) and 1992 (43 days). The major parent-year (1989) spawning escapement was approximately 99,000 fall chums.

Similarly, the interim Fishing Branch River escapement goal was not reached in 1993. A weir passage of only 28,798 fall chums for the period August 31 through October 25 was 42% below the minimum goal of 50,000 fish. The major parent-year escapement (1989) was approximately 43,800 fish.

Tanana River fall chum salmon escapement was evaluated by observations made in the Toklat and Delta River index areas. Escapement to the Toklat River in 1993 was estimated at 28,171 fall chum salmon, being only 15% below the minimum goal of 33,000 fish. By comparison, preliminary information from replicate ground surveys of the Delta River reveals the minimum escapement goal of 11,000 chum salmon was reached. Although these surveys are still being conducted and a final population estimate will not be available prior to mid-December, a peak count of approximately 20,000 spawners was made on November 12. While no escapement goals exist for other fall chum salmon spawning areas in the upper Tanana River, escapement counts during peak spawning were approximately 5,500 and 2,500 fish, respectively, in Bluff Cabin and Clearwater Lake Outlet Sloughs (Big Delta region). These numbers approximate or exceed the averages observed to these areas during the last decade.

The preliminary fall chum spawning escapement estimated by DFO in 1993 for upper mainstem Yukon River stocks is 30,000 fish. An escapement level of this magnitude is the lowest on record since inception of the DFO mark and recapture program in 1982. This upper Yukon River spawning escapement estimate, as part of the twelve year rebuilding plan for the weak 1989 brood year, is 39% below the minimum 1993 targeted level of 51,000 fall chum salmon.

Consistent with other years, I estimated 1993 drainage-wide spawning escapement as the 4-area escapement index, doubled. The doubling factor is an attempt to account for spawning populations in the upper mainstem Yukon River (Yukon Territory), Chandalar River, upper Tanana River, and other areas where spawning occurs but is not monitored. That estimate of escapement totaled 239,938 fall chums. Summing the total inriver catch (113,417) and escapement estimates resulted in an estimated 1993 total return of approximately 353,000 fall chum salmon. This measure of total return was the smallest on record (dating back to 1974) and was 381,000 fish, or 52% lower than the preseason projected return of 734,000. For comparison, the sonar passage estimate at Pilot Station from July 19 through August 31, 1993 was approximately 295,000 fall chums, the lowest sonar passage estimate on record and 40% of the preseason projected return of 734,000 fall chums. The 95% confidence interval for passage at Pilot Station, including an estimate for passage after project termination, was 275,000 to 341,000 fall chums. The major parent year (1989) passage estimate at Pilot Station was 606,000 fall chums.

The poor escapements in 1993 are particularly discouraging in view of the complete closures implemented to all fisheries during the fall season. However, based upon the estimated low percentage of age-4 fish which returned in 1993, production from the winter of 1989-90 (the 1989 brood year) was disastrous.

1994 RUN PROJECTION

Annual inseason estimates of fall chum salmon age composition are shown in Table 3. Whereas inseason estimates of age composition prior to 1981 are based upon fish sampled at Emmonak from 6" commercial gillnet catches, estimates since 1981 are based upon 6" set gillnet test fish catches at Big Eddy and Middle Mouth. On the average, age-4 fish are the most abundant followed by age-5 and age-3 fish. Age-6 fish generally comprise less than 1% of the run annually. Estimated average annual age composition for the period 1977-1993 is approximately 5%, 70%, and 24% for 3, 4, and 5-year-old fish, respectively. Similarly, estimated age composition of brood year returns as shown in Table 4 has averaged approximately 5%, 72% and 23% for age 3, 4, and 5-year-old fish.

Although the overall proportion of age-4 and age-5 fish combined varies little among BYR's, (averaging approximately 95% annually), there is a change in the proportion of these age groups between even and odd brood year returns (see Table 4). For example, age-4 fish average approximately 76% for odd brood year returns, but only 69% for even brood year returns. By comparison, age-5 fish average approximately 20% for odd years and 25% for even years. Since using the overall average maturity schedule would tend to result in a rather conservative projection for the larger in magnitude odd-year returns, while a more robust projection is likely to result for the smaller even-year returns, I chose to use two maturity schedules in making the

1994 run projection: first, the average age composition observed from even brood year returns, and second, the average observed from odd brood year returns.

Table 4 also presents Yukon River fall chum salmon estimated brood year production and return per spawner for the years 1974 through 1988. Results show that fall chum escapements from 1974-1988 resulted in brood year returns ranging in size from approximately 300,000 to 1,400,000 fish (Figure 3). Return per spawner (R/P) rates ranged from 1.14 to 4.56 with smaller escapements tending to result in a higher R/P. Fall chum escapement in even years averaged approximately 224,000 with corresponding BYR's from even-year escapements averaging approximately 507,000 (R/P = 2.27) (Figure 4). By comparison, escapement in odd years averaged approximately 562,000 with corresponding average BYR's of 1,052,000 (R/P = 1.87).

Using logarithms, I regressed the spawner return rate ($\ln R/P$) against the number of spawners (P) for the years 1974-1988 in order to predict specific R/P rates for given levels of escapement. The R/S for any given brood year escapement was then estimated in making the 1994 projection using $R/S = e^{a_2} e^{(bP)}$ (Table 5). The resulting projection is weighted by size of the brood year escapements contributing to the 1994 return in that the predicted R/P rates used in the projection were derived from actual trends in the existing database. This method resulted in a 1994 projection of 717,096 fish with the following age composition:²

Age-3 fish	31,257 (1991 Brood Year)
Age-4 fish	510,583 (1990 Brood Year)
Age-5 fish	172,222 (1989 Brood Year)
Age-6 fish	3,034 (1988 Brood Year)

As can be seen, the suspected major contributor to the 1994 return will be from the brood year 1990. In that year, the total inriver return was estimated as approximately 665,000 fall chum salmon, with escapement goals being met in only the Toklat and Sheenjek Rivers. Escapement goals were not achieved in either the Delta or Fishing Branch Rivers in 1990, nor for upper Yukon River mainstem Canadian stocks. Although weaknesses in the 1994 return can likely be expected in age-4 fish from the 1990 brood year among the Canadian and upper Tanana River stocks, it should be stressed that production of age-5 chum salmon from the 1989 brood year is expected to be very low. This is based upon the widespread failure in production of age-4 fish from the 1989 brood year manifest in the 1993 run.

The preseason 1993 projection of 734,000 fall chum salmon included an estimated return of approximately 623,000 age-4 fish from the 1989 parent year. However, it was estimated that only about 219,000 of those fish actually returned in 1993; 65% lower than projected.

²If the average maturity schedule were used, the 1994 projection would be 767,548 fall chum salmon. This information is also shown in Table 5.

Assuming that 1993 typified recruitment from the 1989 brood year, the final 1994 projection includes a 112,000 age-5 shortfall (i.e., 65% of the 172,000 age-5 projected return) and becomes 605,151 fall chum salmon. This represents a R/S rate of only 0.8 for the 1989 brood year, age-5 fish.

A word of caution: Apart from the anticipated age-5 shortfall next season, the 1994 projection includes approximately 511,000 age-4 recruitment from the brood year 1990. This is based upon a predicted R/S rate of 2.4 (under normal conditions), given an escapement estimate of approximately 311,000 spawners in 1990 and an average of 69% age-4 fish returning from even-numbered parent years. However, productivity of age-4 fish returning in 1992 and 1993 from the parent years 1988 and 1989, was estimated at only 1.2 and 0.8. If the poor production levels observed for the brood years 1988 and 1989 carries over to 1990 (i.e., a recruitment of near 1:1), then the 1994 projected return of age-4 fish would only be 214,000 chum salmon. If such should occur, the 1994 projection of approximately 605,000 fall chum salmon would be high by some 296,000 fish. This would result in the most disastrous run outlook on record (309,000). That production from 1990 could be extremely poor is supported in a qualitative way based upon estimated age composition of the fall chum run in 1993. In that year, the lowest proportion of age-3 fish on record (<2%) was observed, with less than 1,000 age-3 fall chums estimated to have returned (see Table 3).

HARVESTABLE SURPLUS

The minimum number of fall chum salmon needed to allow for prosecution of a commercial fishery is dependent upon, and would fluctuate in proportion to, the sum of the number of fish needed to meet both escapement requirements and subsistence needs throughout the drainage. Available U.S. commercial harvest would also be influenced by the number of fish the U.S. would endeavor to deliver to the Canadian border as tentatively agreed to with respect to the 12-year rebuilding plan for Yukon River mainstem fall chum salmon.

The most recent four-year average U.S. subsistence harvest of fall chum salmon is 142,200 fish. However, this number is likely a low estimate of the annual U.S. subsistence need, since it includes the low harvest estimates for 1992 (the year of early freeze-up) and 1993 (the year of a very limited subsistence fishery). Consequently, I consider it more reasonable to estimate the U.S. subsistence need as the average harvest estimated since 1980. That harvest averaged approximately 182,000 fall chum salmon (See Table 1). The guideline harvest range for Canadian Yukon mainstem fisheries (commercial plus subsistence) is 23,600 to 32,600, with a targeted harvest toward the low end of the range in years of low returns and the upper end of the range in years of strong returns. Thus, I consider 206,000 to be a good approximation of the number of fall chum salmon needed to meet U.S. subsistence needs and our commitment to Canada in years of anticipated weak returns, such as in 1994.

Prior to the 1993 fishing season, we identified 400,000 fall chum salmon as a drainage-wide escapement target in an attempt to help insure individual escapement goals were reached in the five areas where goals exist. For the moment, considering 206,000 fall chum salmon to approximate the annual drainage-wide subsistence and Canadian need, together with an inriver escapement need of 400,000, would suggest that a total return size in excess of 606,000 fish are needed before consideration is given to allowing for prosecution of a U.S. commercial fishery. Using this scenario and a 1994 projection of 605,000 fall chum salmon would indicate only 205,000 fish would be potentially available for harvest. Such a surplus would indicate no commercial fishing opportunities would exist in 1994, and is slightly below anticipated subsistence needs. Further, such a surplus would suggest subsistence fishery restrictions may be needed, particularly if run distribution is poor. However, poor production from 1990 (e.g., a return rate of near 1:1) would exacerbate the situation even more.

As a final step, I submitted the fall chum salmon data to the Ricker model $R = \alpha P e^{(-BP)}$, in a program recently developed by Jeff Bromaghin, to obtain an estimate of the aggregate number of spawners needed to produce an inriver MSY. His program (RPSEM) fits the best possible curve through observed spawner, return pairs. The model parameters are estimated such that the sum of the absolute residuals (the difference between observed and predicted returns) is minimized. Results indicated that approximately 548,000 spawners (P_s) are needed to produce an MSY of 544,000 fish. These results suggest that targeting an escapement of 400,000 fish may likely be a level well below that which produces MSY for Yukon River fall chum salmon.

As a matter of interest, Figure 5 shows previous Yukon River fall chum salmon projections versus the estimated total return for years in which projections were made (1987-1992).

Distribution: Bergstrom, Blaney, Bromaghin, Buklis, Cannon, Hilsinger, Holder, Kron, Sandone, Schneiderhan, Schultz

cc: Andersen (SF)

Table 1. Yukon River fall chum salmon estimated harvest and escapement data, 1974–1993.

Year	Alaska ^a			Canada ^a			Entire River			Estimated Escapement ^b	Estimated Return	Exploitation Rate ^c
	Comm ^b	Subs	Total	Comm	Subs ^c	Total	Comm	Subs	Total			
1974	289,778	93,778 ^d	383,552	2,544	9,102	11,646	292,320	102,878	395,198	343,780	738,978	0.5348
1975	275,009	86,591 ^d	361,600	2,500	18,100	20,600	277,509	104,691	382,200	1,242,742	1,624,942	0.2352
1976	156,360	72,327 ^d	228,717	1,000	4,200	5,200	157,390	76,527	233,917	246,264	480,181	0.4871
1977	257,986	82,771	340,757	3,990	8,489	12,479	261,976	91,260	353,236	374,564	727,800	0.4853
1978	236,383	94,867	331,250	3,356	6,210	9,566	239,739	101,077	340,816	242,884	583,700	0.5839
1979	359,946	233,347	593,293	9,084	13,000	22,084	369,030	246,347	615,377	798,504	1,413,881	0.4352
1980	293,430	172,657	466,087	9,000	13,218	22,218	302,430	185,875	488,305	231,422	719,727	0.6785
1981	466,451	188,525	654,976	15,260	7,021	22,281	481,711	185,546	677,257	342,468	1,019,715	0.6642
1982	224,187	132,897	357,084	11,312	4,779	16,091	235,499	137,676	373,175	110,316	483,491	0.7718
1983	302,598	192,928	495,526	25,990	3,500	29,490	328,588	186,428	525,016	210,208	735,224	0.7141
1984	208,232	174,823	383,055	22,932	6,335	29,267	231,164	181,158	412,322	142,404	554,726	0.7433
1985	267,744	206,472	474,216	35,746	5,519	41,265	303,490	211,991	515,481	497,730	1,013,211	0.5088
1986	139,442	184,043	303,485	11,484	3,029	14,493	150,906	167,072	317,978	281,052	599,030	0.5308
1987	0	361,663 ^a	361,663	40,591	3,889	44,480	40,591	365,552 ^a	406,143	464,726	870,869	0.4664
1988	160,963 ^f	159,703	320,666	30,263	3,302	33,565	191,226	163,005	354,231	191,622	545,853	0.6489
1989	287,179 ^f	224,046	511,225	17,549	5,471	23,020	304,728	229,517	534,245	389,478	923,723	0.5784
1990	132,118 ^f	188,941	321,059	27,537	6,085	33,622	159,655	195,026	354,681	310,728	665,409	0.5330
1991	234,788 ^f	168,890 ^g	403,678	31,404	4,014	35,418	266,192	172,904	439,096	340,662	779,758	0.5631
1992 [*]	17,128 ^h	111,109 ^k	128,237	18,576	2,044	20,620	35,704	113,153	148,857	246,824	395,681	0.3762
1993 [*]	0	100,000 ^a	100,000	7,762	5,655	13,417	7,762	105,655	113,417	239,936	353,355	0.3210
Averages:												
1974–1977	244,790	83,866	328,657	2,509	9,973	12,481	247,299	93,839	341,138	551,838	892,975	0.4356
1978–1981	339,053	172,349	511,402	9,175	9,882	19,037	348,228	182,211	530,439	403,817	934,256	0.5904
1982–1985	250,690	176,780	427,470	23,995	5,033	29,028	274,685	181,813	456,499	240,165	696,663	0.6845
1986–1989	146,896	227,364 ^e	374,260	24,967	3,923	28,890	171,863	231,287	403,149 ✓	331,720 ✓	734,869	0.5561
1990–1993	96,009	142,235	238,244	21,320	4,450	25,769	117,328	146,685	264,013 ✓	284,538 ✓	548,551 ✓	0.4483 ✓
1974–1993	215,488	160,519	376,006	16,393	6,648	23,041	231,881	167,167	399,047	362,415	761,463	0.5430
Eve yr 74–93	185,805	136,514	322,319	13,798	5,830	19,629	199,603	142,345	341,948	234,730	576,678	0.5888
Odd yr 74–93	246,883	185,634	432,517	20,069	7,105	27,174	266,952	192,739	459,691	490,790	950,480	0.5027
1980–1993	195,304	181,907	377,211	21,813	5,276	27,089	217,118	187,183	404,300	285,683	689,984	0.5785
Eve yr 80–93	167,929	157,739	325,668	18,726	5,542	24,268	186,655	163,281	349,936	216,338	566,274	0.6118
Odd yr 80–93	222,680	206,075	428,755	24,900	5,010	29,910	247,580	211,085	458,665	355,029	813,694	0.5451

^{*} Preliminary data.^a All U.S. and Canadian harvest data are from 1974–77 AMR's and Appendix A.20 of 1991 AMR unless otherwise noted.^b Alaskan commercial harvest does not include pounds of roe sold, as females that provided roe are included in subsistence harvest estimates.^c Canadian subsistence includes the Indian food and domestic fisheries.^d U.S. subsistence fall chum salmon catch in these years was estimated from average percentages calculated from 1977 and 1978 data: i.e., the average percent of total summer and fall chum subsistence caught fish which were fall chums. Data from 1979–1981 were not included in calculations since reported fall chum salmon subsistence utilization increased dramatically in those years. (See Table 13 in 1977 and 1978 AMR's).^e Although no commercial fall chum salmon fishery occurred in 1987, this figure includes an estimated 115,829 fall chum salmon not documented by the department primarily due to an estimated illegal sales from districts 5 and 6 of 42,274 chum salmon and a documented 110,948 pounds of illegally processed roe. Data from Regnart memo dtd November 6, 1988. The 115,829 fish estimate derived as follows: an estimated 107,467 pounds of unprocessed fall chum salmon roe times 1.0 lbs/female times 2 (assume 1:1 sex ratio) minus reported subsistence harvest in Tanana, Ramparts, Fairbanks "Fish Camp", Manley, and Nenana (99,107 fish). Assumes that salmon illegally sold in the round were from these fish which provided roe.^f Includes ADF&G test fish sales as follows: 26,988 in 1988; 16,984 in 1989; 7,060 in 1990; 3,936 in 1991; and 1,407 in 1992^g Data from Table 10, draft T.F.R., Bromaghin and Hamner, November 1992 (estimated Yukon River subsistence harvests in 1991).^h Total escapement estimates taken as twice the 4–area index (Delta, Toklat, Sheenjek, and Fishing Branch River) estimates.ⁱ Sum of entire river commercial and subsistence harvest divided by estimated total return. This is considered a maximum exploitation rate as annual estimated total escapement figures are thought to be on the conservative side.^j Data from Appendix F.3 of 1992 AMR.^k Data from Appendices A.3 and F.3 1992 AMR.^a Subjectively estimated as approaching 1992 harvest level; the year of early freezeup.

Table 2. Yukon River fall chum salmon spawning escapement population estimates, 1974–1993.

Year	Escapement to Four Index Areas					Yukon River Esc Index (4–Area Indx doubled)	Relative Contribution of Each Component Stock				Canada Mainstem Spawning Escapement ⁱ
	Delta ^a	Toklat ^b	Sheenjek ^c	Fish Br ^d	4–Area Total		Delta	Toklat	Sheenjek	Fish Br	
1974	5,915	43,484	89,966	32,525 ^w	171,890	343,780	0.03	0.25	0.52	0.19	
1975	3,734 ^p	90,984	173,371	353,282 ^w	621,371	1,242,742	0.01	0.15	0.28	0.57	
1976	6,312 ^p	53,882	26,354	36,584	123,132	246,264	0.05	0.44	0.21	0.30	
1977	16,876 ^p	36,462	45,544	88,400	187,282	374,564	0.09	0.19	0.24	0.47	
1978	11,136	37,057	32,449	40,800	121,442	242,884	0.09	0.31	0.27	0.34	
1979	8,355	179,627	91,372	119,898	399,252	798,504	0.02	0.45	0.23	0.30	
1980	5,137	26,373	28,933	55,268	115,711	231,422	0.04	0.23	0.25	0.48	
1981	23,508	15,775	74,560 ^a	57,386 ^c	171,229	342,458	0.14	0.09	0.44	0.34	
1982	4,235	3,601	31,421 ^a	15,901	55,158	110,316	0.08	0.07	0.57	0.29	31,958
1983	7,705	20,807	49,392 ^a	27,200	105,104	210,208	0.07	0.20	0.47	0.26	90,875
1984	12,411	16,511	27,130 ^a	15,150	71,202	142,404	0.17	0.23	0.38	0.21	56,633 ^j
1985	17,276 ^p	22,805	152,768 ^a	56,016 ^w	248,865	497,730	0.07	0.09	0.61	0.23	62,010
1986	6,703 ^p	18,903	83,197 ^a	31,723 ^w	140,526	281,052	0.05	0.13	0.59	0.23	87,990
1987	21,180	22,141	140,086 ^a	48,956 ^w	232,363	464,726	0.09	0.10	0.60	0.21	80,776
1988	18,024	13,324	40,866 ^a	23,597 ^w	95,811	191,622	0.19	0.14	0.43	0.25	36,786
1989	21,342 ^p	30,447	99,116 ^{a,f}	43,834 ^w	194,739	389,478	0.11	0.16	0.51	0.23	35,750
1990	8,992 ^p	33,672	77,700 ^{a,g}	35,000 ^h	155,364	310,728	0.06	0.22	0.50	0.23	51,735
1991	32,905 ^p	13,197	86,496 ^a	37,733 ^w	170,331	340,662	0.19	0.08	0.51	0.22	78,461
1992	8,893 ^p	13,194	78,808 ^a	22,517 ^w	123,412	246,824	0.07	0.11	0.64	0.18	49,277
1993 [*]	20,000	28,171	43,000 ^a	28,798 ^w	119,969	239,938	0.17	0.23	0.36	0.24	29,938
Average	13,032	36,021	73,626	58,528	181,208	362,415	0.09	0.19	0.43	0.29	57,682
Escapement Objectives	>11,000	>33,000	>64,000	50,000 – 120,000	>158,000						>80,000

^{*} Preliminary data.

^a Total escapement estimates made from migratory time density curve (Barton 1986) unless otherwise indicated; (^p) population estimate from replicate foot surveys and stream life data.

^b Total escapement estimates using Delta River migratory time density curve and percentage of live salmon present by survey date in the upper Toklat River area.

^c Total escapement estimates using sonar to aerial survey expansion factor of 2.22 unless otherwise indicated; (^s) sonar estimate.

^d Total escapement estimates using weir to aerial survey expansion factor of 2.72 unless otherwise indicated; (^w) weir estimate.

^e Initial aerial survey count was doubled before applying the weir/aerial expansion factor of 2.72 since only half of the spawning area was surveyed.

^f Includes an estimated 20,000 fall chum salmon present in river prior to sonar operations. Sonar count was 79,116 chum salmon.

^g Includes an estimated 15,500 fall chum salmon present in river prior to sonar operations. Sonar count was 62,200 chum salmon.

^h Weir was not operated. DFO estimated population of spawners to be approximately 30,000–40,000 fish, based upon a late October 26 aerial survey (7,541 fish actually observed).

ⁱ Estimated border passage minus Canadian removal (excludes Fishing Branch River escapement).

^j Estimate based upon assumed average exploitation rate.

Table 3. Yukon River fall chum salmon estimated total annual return and age composition of annual returns based upon 6" gill net samples in lower river.^a

Year	(P)	(C)	(R)	Annual Age Composition Estimates							
	Estimated Annual Totals			Percent				Number			
	Escapement	Catch	Return	Age 3	Age 4	Age 5	Age 6	Age 3	Age 4	Age 5	Age 6
1974	343,780	395,198	738,978								
1975	1,242,742	382,200	1,624,942								
1976	246,264	233,917	480,181								
1977	374,564	353,236	727,800	0.0953	0.8505	0.0531	0.0011	69,389	618,985	38,637	789
1978	242,884	340,816	583,700	0.1994	0.6597	0.1394	0.0015	116,390	385,049	81,385	875
1979	798,504	615,377	1,413,881	0.0731	0.8777	0.0492	0.0000	103,411	1,240,935	69,535	0
1980	231,422	488,305	719,727	0.1366	0.7817	0.0817	0.0000	98,304	562,616	58,807	0
1981	342,458	677,257	1,019,715	0.0164	0.8859	0.0976	0.0000	16,747	903,415	99,552	0
1982	110,316	373,175	483,491	0.0859	0.5664	0.3359	0.0117	41,532	273,849	162,405	5,657
1983	210,208	525,016	735,224	0.0092	0.8962	0.0901	0.0046	6,735	658,895	66,226	3,367
1984	142,404	412,322	554,726	0.0739	0.5317	0.3945	0.0000	40,982	294,927	218,817	0
1985	497,730	515,481	1,013,211	0.0143	0.8379	0.1455	0.0022	14,522	848,997	147,457	2,234
1986	281,052	317,978	599,030	0.0207	0.5745	0.4007	0.0041	12,394	344,132	240,025	2,479
1987	464,726	406,143	870,869	0.0067	0.8184	0.1631	0.0111	5,835	712,719	142,039	9,667
1988	191,622	354,231	545,853	0.0154	0.6006	0.3793	0.0046	8,417	327,848	207,062	2,525
1989	389,478	534,245	923,723	0.0020	0.8381	0.1548	0.0050	1,847	774,172	142,992	4,619
1990	310,728	354,681	665,409	0.0100	0.5970	0.3800	0.0130	6,654	397,249	252,855	8,650
1991	340,662	439,096	779,758	0.0023	0.6408	0.3538	0.0031	1,793	489,669	275,878	2,417
1992	246,824	148,857	395,681	0.0062	0.3944	0.5901	0.0093	2,453	156,057	233,491	3,680
1993	239,938	113,417	353,355	0.0018	0.6188	0.3620	0.0173	636	218,656	127,915	6,113
Average	362,415	399,047	761,463	0.0453	0.7041	0.2453	0.0052	32,238	542,245	150,887	3,122
Even Year Avgs (1978–1992)				0.0685	0.5882	0.3377	0.0055	40,891	342,716	181,856	2,983
Odd Year Avgs (1977–1993)				0.0246	0.8071	0.1632	0.0049	24,546	719,605	123,359	3,245
Even Year Average (1981–93)				0.0353	0.5441	0.4134	0.0071	18,739	299,010	219,109	3,832
Odd Year Average (1981–93)				0.0075	0.7909	0.1953	0.0062	6,874	659,503	143,151	4,060

^a Annual age composition estimates are: District 1 commercial gill net samples (6" mesh) for years 1977–1980, and ADF&G test set gillnet samples (=6") at Big Eddy and Middle Mouth for years 1981–1993.

Table 4. Yukon River fall chum salmon estimated brood year production and return per spawner (1978-88).

Year	(P)	(C)	(R)	Estimated Brood Year Return (BYR)								(BYR)	(R/P)
	Estimated Annual Totals			Number of Salmon *				Percent				Total Brood Year Return *	Return/Spawner (R/P)
	Escapement	Catch	Return	Age 3	Age 4	Age 5	Age 6	Age 3	Age 4	Age 5	Age 6		
1974	343,780	395,198	738,978	69,359	385,067	69,563	0	0.132	0.735	0.133	0.000	523,989	1.52
1975	1,242,742	382,200	1,624,942	116,390	1,240,963	58,802	0	0.082	0.878	0.042	0.000	1,416,155	1.14
1976	246,264	233,917	480,181	103,355	562,611	99,524	5,657	0.134	0.730	0.129	0.007	771,146	3.13
1977	374,564	353,236	727,800	98,315	903,366	162,405	3,382	0.084	0.774	0.139	0.003	1,167,467	3.12
1978	242,884	340,816	583,700	16,723	273,849	66,244	0	0.047	0.767	0.186	0.000	356,816	1.47
1979	798,504	615,377	1,413,881	41,532	658,908	218,839	2,229	0.045	0.715	0.237	0.002	921,508	1.15
1980	231,422	488,305	719,727	6,764	294,948	147,422	2,456	0.015	0.653	0.326	0.005	451,590	1.95
1981	342,458	677,257	1,019,715	40,994	848,669	240,031	9,667	0.036	0.745	0.211	0.008	1,139,662	3.33
1982	110,316	373,175	483,491	14,489	344,143	142,039	2,511	0.029	0.684	0.282	0.005	503,181	4.56
1983	210,208	525,016	735,224	12,400	712,719	207,042	4,619	0.013	0.761	0.221	0.005	936,780	4.46
1984	142,404	412,322	554,726	5,835	327,839	142,992	8,650	0.012	0.676	0.295	0.018	485,317	3.41
1985	497,730	515,481	1,013,211	8,406	774,172	252,855	2,417	0.008	0.746	0.244	0.002	1,037,851	2.09
1986	281,052	317,978	599,030	1,847	397,249	275,878	3,680	0.003	0.585	0.407	0.005	678,655	2.41
1987	464,726	406,143	870,869	6,654	499,669	233,491	6,113	0.009	0.670	0.313	0.008	745,927	1.61
1988	191,622	354,231	545,853	1,793	156,057	127,915						285,765 ^b	1.49
1989	389,478	534,245	923,723	2,453	218,656								
1990	310,728	354,681	665,409	636									
1991	340,662	439,096	779,758										
1992	246,824	148,857	395,681										
1993	239,938	113,417	353,355										
Average	362,415	399,047	761,463					0.0464	0.7226	0.2260	0.0050	761,454	2.46
	223,718							0.0531	0.6900	0.2510	0.0059	507,057	2.27
	561,562							0.0397	0.7552	0.2009	0.0042	1,052,193	1.87

* The estimated numbers of salmon which returned are based upon the actual age compositions observed in lower Yukon test nets each year (see Table 3).

^b Brood year return includes only 3, 4, and 5 year fish.

Table 5. Yukon River fall chum salmon projected returns based upon predicted return per spawner rates and brood year return (BYR) maturity schedules.

Year	(P)	(R/P)	(R/P) ^a	Previous Run Projections	Observed Returns	Using Predicted R/P	
	Estimated Total Escapement	Observed Return/ Spawner	Predicted Return/ Spawner			Projections using odd/even maturity sch	Projections using average maturity sch
1974	343,780	1.52	2.30		738,978		
1975	1,242,742	1.14	0.94		1,624,942		
1976	246,264	3.13	2.54		480,181		
1977	374,564	3.12	2.23		727,800		
1978	242,884	1.47	2.55		583,700		
1979	798,504	1.15	1.46		1,413,881		
1980	231,422	1.95	2.58		719,727	703,554	758,081
1981	342,458	3.33	2.31		1,019,715	826,735	780,601
1982	110,316	4.56	2.91		483,491	645,281	693,724
1983	210,208	4.46	2.63		735,224	1,072,173	1,015,313
1984	142,404	3.41	2.82		554,726	681,208	734,695
1985	497,730	2.09	1.98		1,013,211	768,353	726,465
1986	281,052	2.41	2.45		599,030	405,727	439,195
1987	464,726	1.61	2.04	404,757	870,869	523,245	495,117
1988	191,622	1.49	2.68	589,415	545,853	429,037	462,312
1989	389,478		2.20	861,000	923,723	882,093	835,835
1990	310,728		2.38	784,155	665,409	713,269	766,416
1991	340,662		2.31	855,121	779,758	920,949	870,126
1992	246,824		2.54	601,355	395,681	583,408	629,102
1993	239,938 *		2.56	734,000 ^b	353,355	819,707	774,710
1994				605,151 ^c		717,096	767,548

BYR Age Composition	Age 3	Age 4	Age 5	Age 6
Average 1974–1987	0.0464	0.7226	0.2260	0.0050
Average Even Years	0.0531	0.6900	0.2510	0.0059
Average Odd Years	0.0397	0.7552	0.2009	0.0042

^a Predicted R/P = $e^{(a)} \cdot e^{(b \cdot P)}$; $a = 1.178523$; $b = -0.0000010$

Therefore, $R/P = e^{(1.178523)} \cdot e^{(0.0000010 \cdot P)}$

^b This final projection included an 80,000 age–5 shortfall from the 1988 brood year, the year of the "omega block".

^c Final projection includes 112,000 age–5 shortfall from the brood year 1989.

* Preliminary.

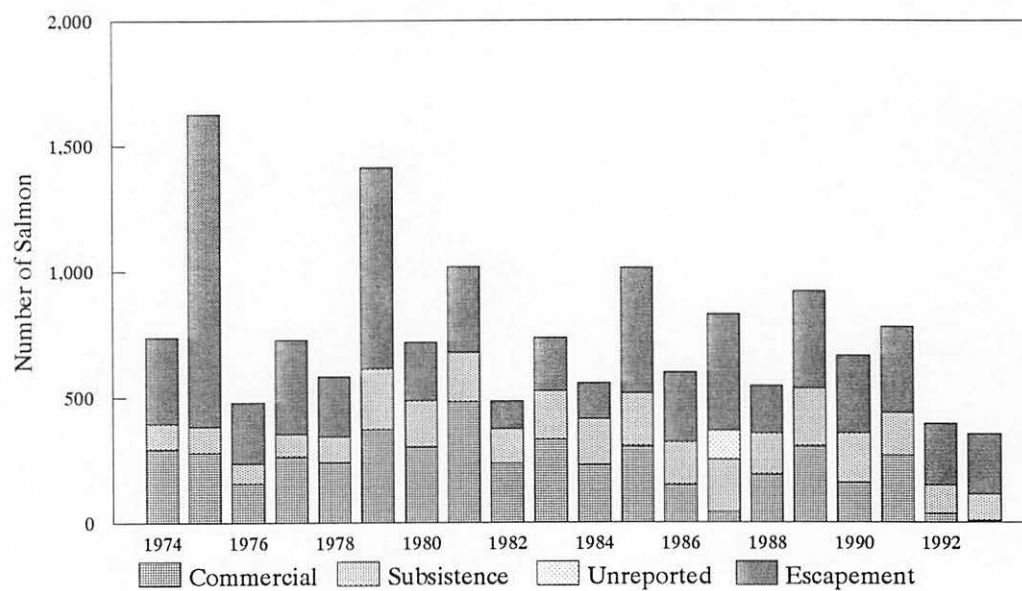


Figure 1. Estimated fall chum salmon harvest and escapement by fishery, 1974-1993.

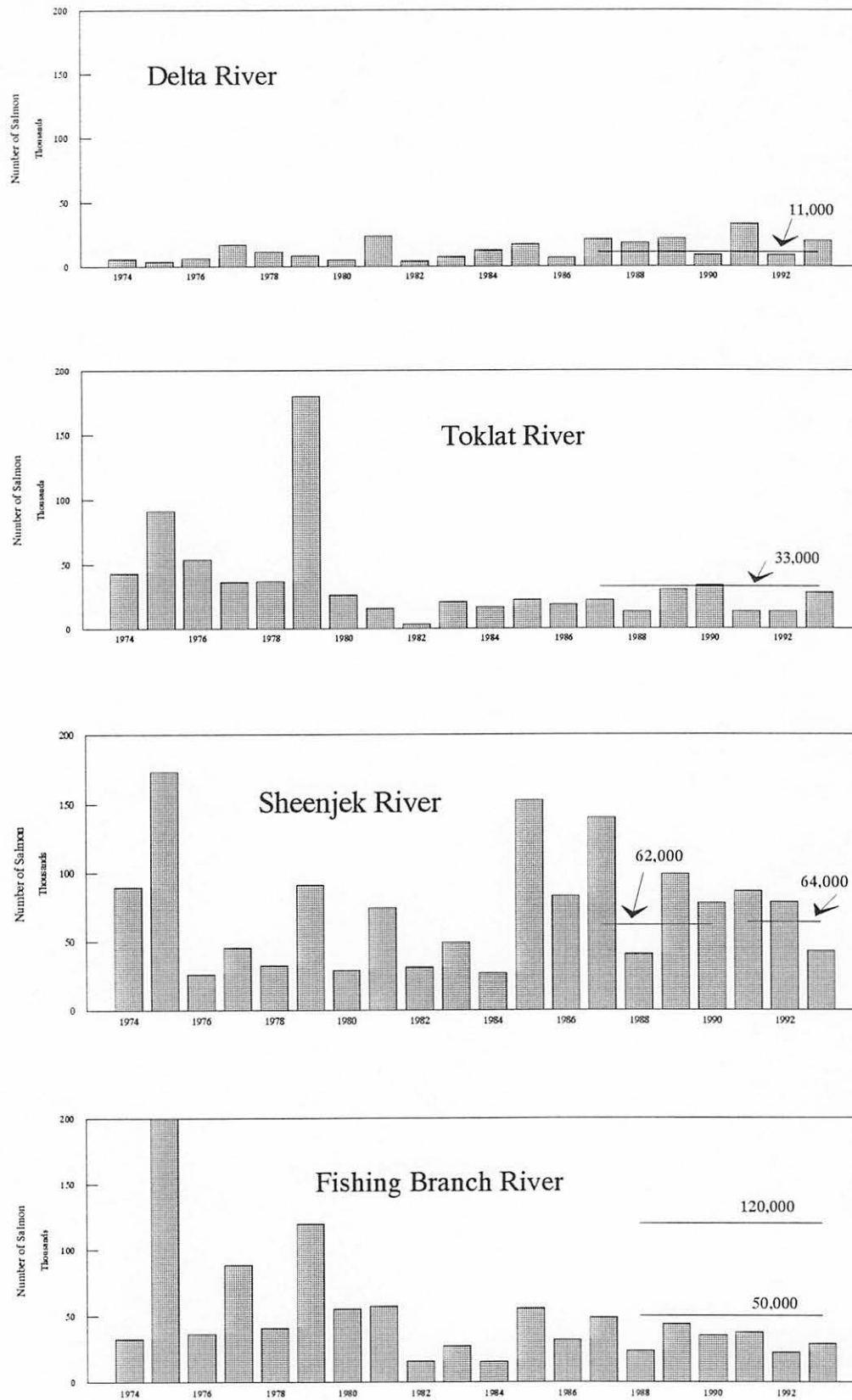


Figure 2. Fall chum escapement estimates for Delta, Toklat, Sheenjek, and Fishing Branch Rivers, 1974–1993.

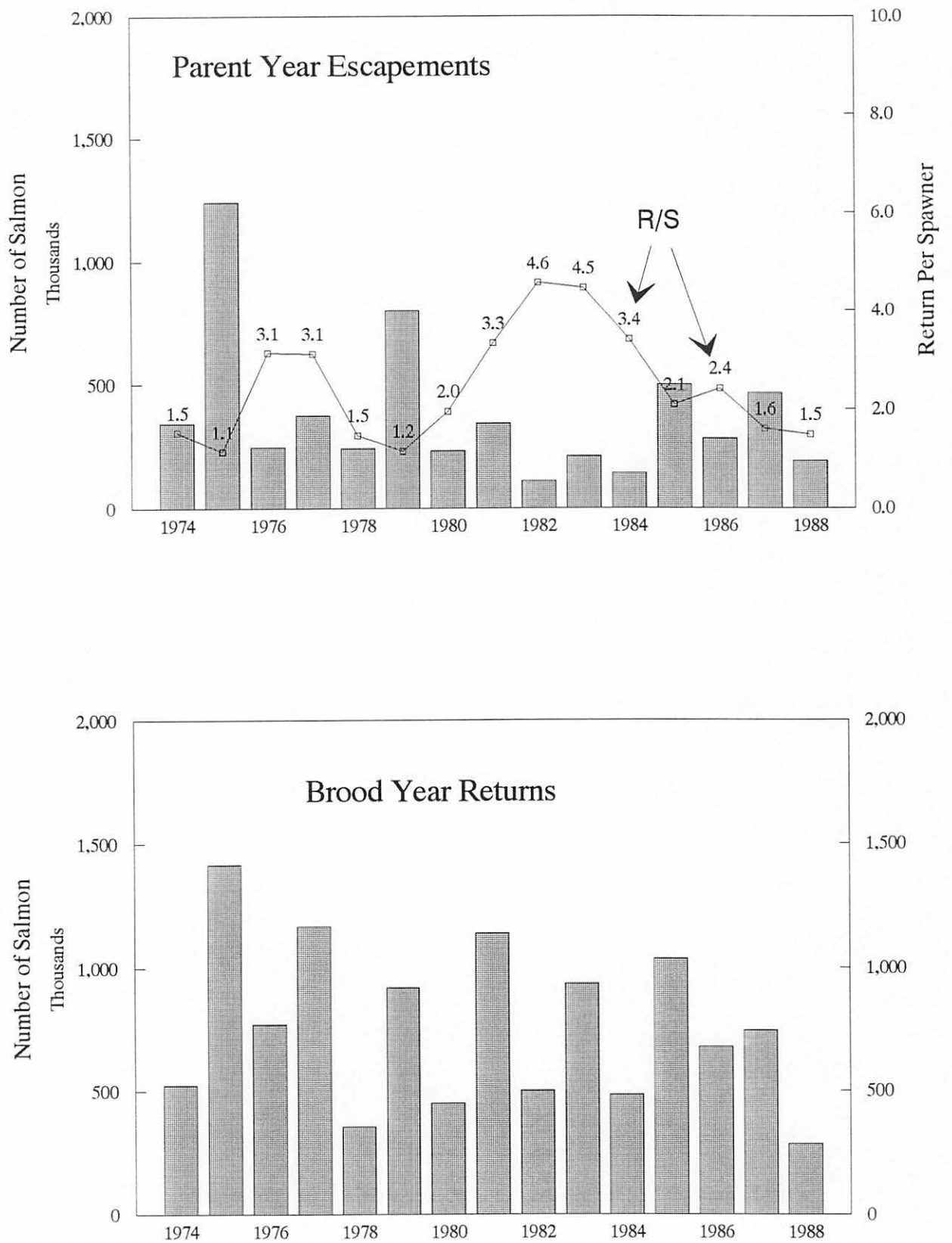


Figure 3. Estimated parent year escapements and brood year returns of Yukon River fall chum salmon, 1974-1988.

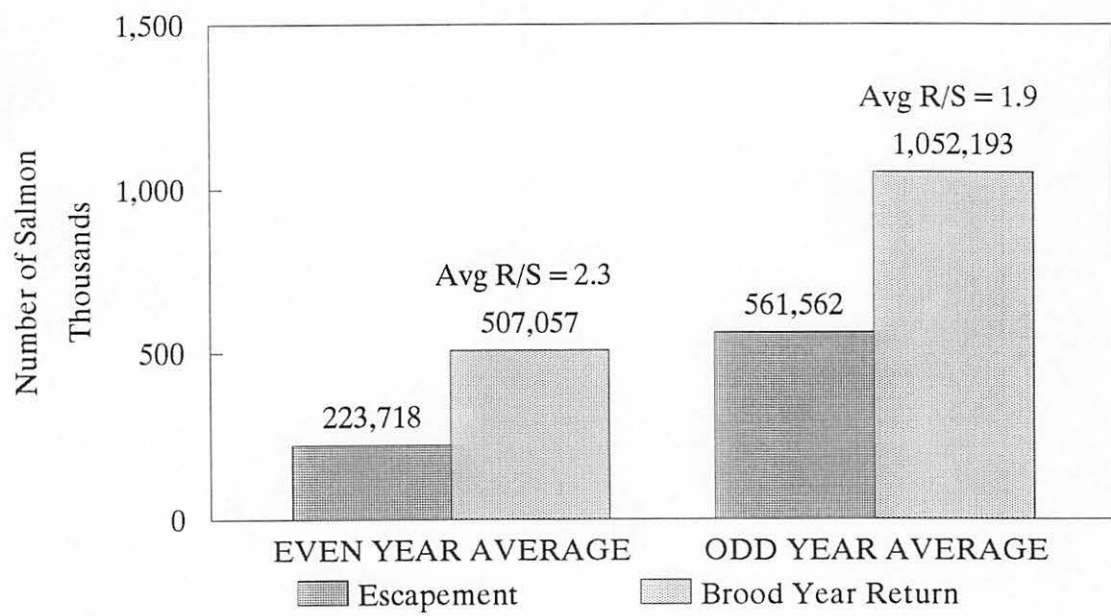


Figure 4. Average even-odd year production of Yukon River fall chum salmon, 1974-1988.

Yukon River Fall Chum Salmon

Projected and Observed Returns

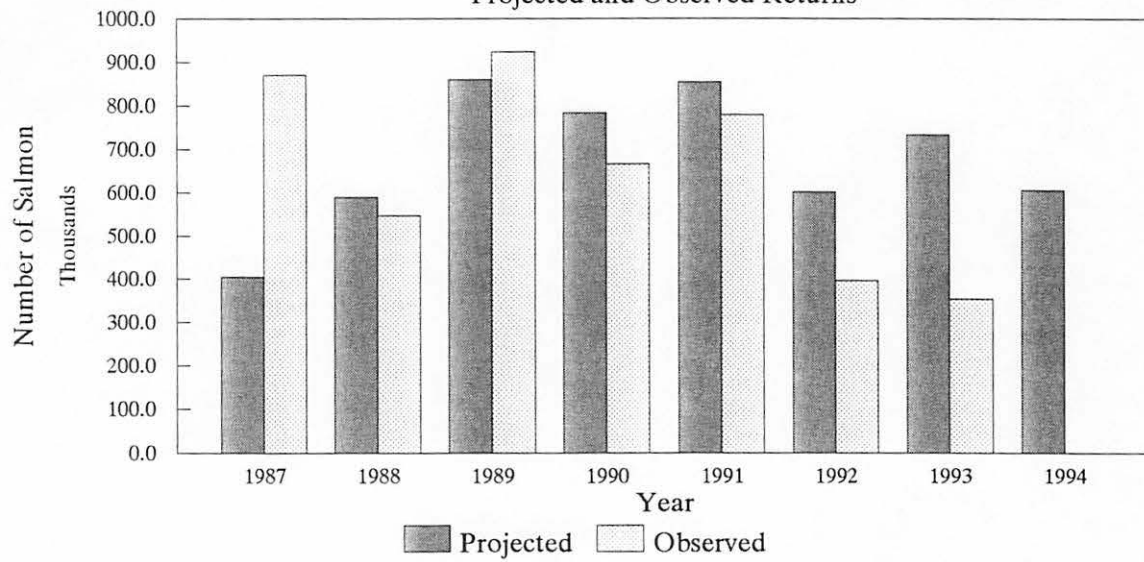


Figure 5. Preseason run projections versus observed returns for Yukon River fall chum salmon, 1987–1994.